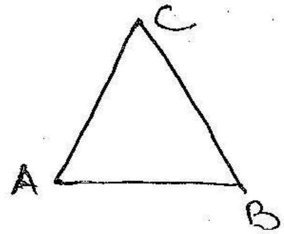
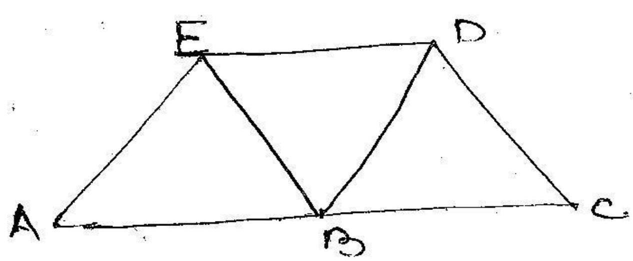
Trusses

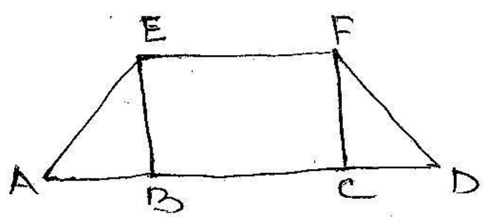
A Structure, made up of several straight members which are rivetted, bolted or welded together to form a net work of triangles is known as truss.
This structure is designed to resist geometrical distortion under any applied system of loading.
Some members are joind to make a truss as show below



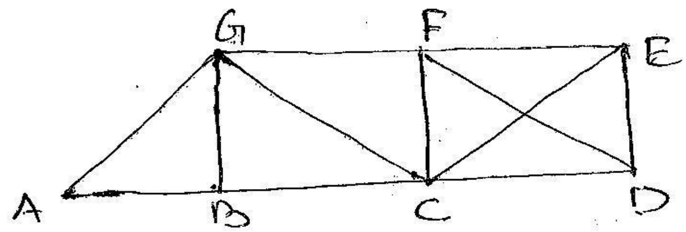
There are 3 members & 3 joints
If the truss satisfy the relation
$M = 2j - 3$ then the truss is
said to be perfect truss
In the given truss
M is the No of members
j is the No of joints
Hence $M = 2j - 3$
$3 = 2\times3 - 3 = 3$ — Perfect



no of mem = 7 No. of joints = 5
$M = 2j - 3$
$7 = 2\times5 - 3 = 7$ Perfect



No. of Members = 8 No of joints = 6
$M = 2j - 3$
$8 \neq 2\times6 - 3 = 9$ Imperfect truss
No. of Member are Less than requir
Hence truss is deficient.

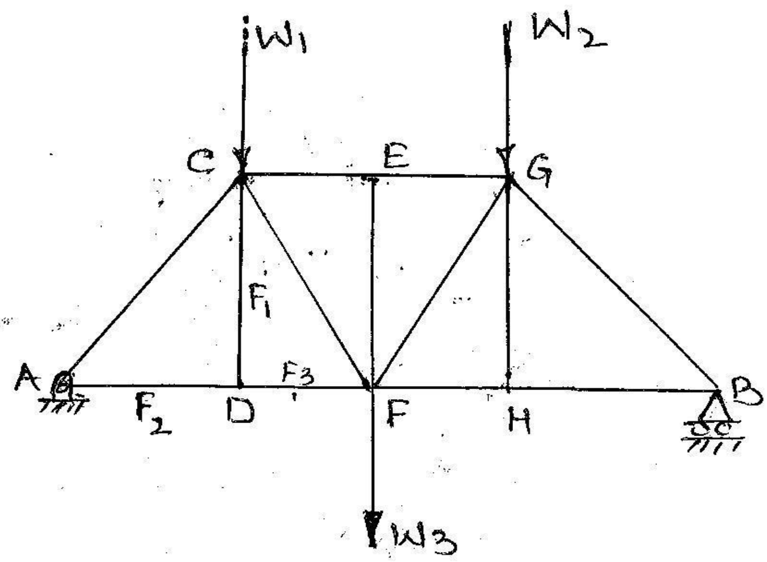


No. of Members = 12
No. of joints = 7

$M = 2j - 3$

$12 \neq 2 \times 7 - 3 = 11$

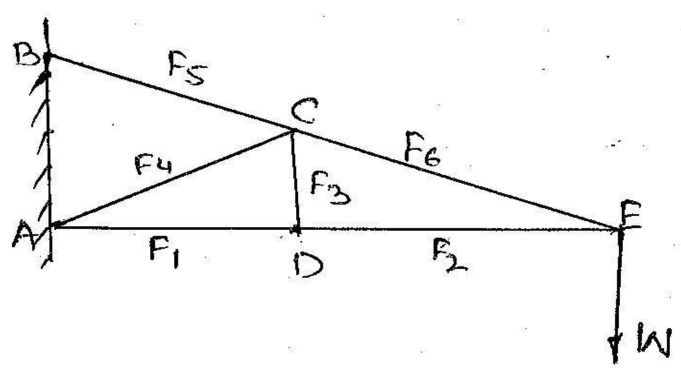
No of Members are more than required
Hence Imperfect truss
Redudant truss.



$M = 2j - 3$

$13 = 2 \times 8 - 3 = 13$ Perfect

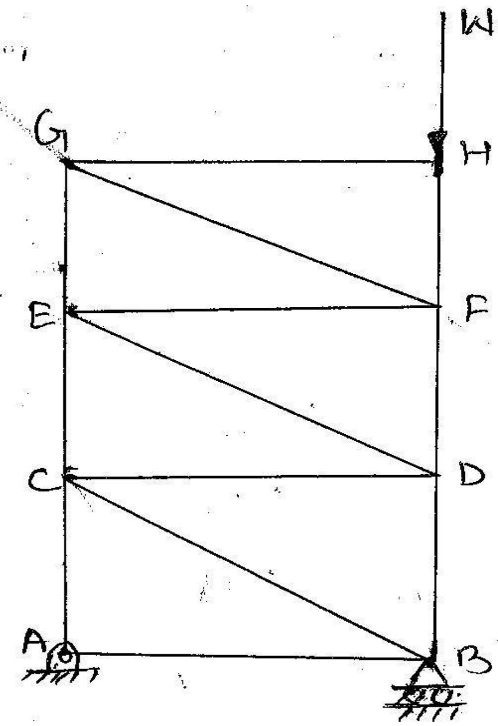
joint D is unloaded 4
F_2 and F_3 are collinear
So F_1 is zero
Forces in the members
EF 4 HG are also zero



$M = 2j - 3$

$7 = 2 \times 5 - 3 = 7$ Perfect

joint D is unloaded
F_3 in member CD is zero
Joint C is also unloaded
(4 F_3 is zero) hence
F_4 in AC is zero.



$M = 2j - 3$

$13 = 8 \times 2 - 3 = 13$ Perfect

Reaction at roller support will be Perpendicular

$R_B \uparrow = W \downarrow$

Hence Reaction at Hinged support will be zero $R_A = 0$

obviously there will No force in members other than BD, DF & FH

TRUSS :

A truss is a structure, made up of several straight members, rivetted or bolted or welded together to form a network of triangles.
This structure is designed to resist geometrical distortion under any applied system of loading.

CLASSIFICATION OF TRUSSES :

Trusses are generally classified as :-

- Plane truss and space truss :

  In plane truss, all the members lie in a single plane and the forces act along the plane of the truss.
  For example: Bridge truss and roof truss.

  A truss in which all the members don't lie in the same plane is called a space truss.
  For example: Tripod and suspention towers.

- Statically determinate and Statically Indeterminate Trusses :

  A truss is statically determinate if the equation of static equilibrium alone are sufficient to determine axial forces in the members without the need of considering their deformations.

  Equations of static equilibrium are not sufficient to determine the forces in statically indeterminate trusses; there is need of considering their deformation also.

- Perfect and Imperfect Truss :

A perfect truss is made up of members just sufficient to keep it in equilibrium when loaded. The no. of members in a perfect truss should satisfy the relation

$$m = 2j - 3$$

where m = no. of members
j = no. of joints.

An imperfect truss is one which does not satisfy the relation $m = 2j - 3$.

The imperfect truss may be further classified into

1. Deficient truss : It is one in which no. of members are less than $2j - 3$.
2. Redundant truss : In this the no. of members are more than $2j - 3$.

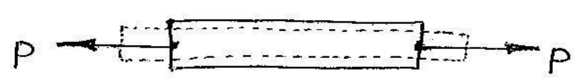
STRESS :

When body is acted upon by a force, the internal force which is transmitted through the body is known as stress.

Following two types of stress are important :

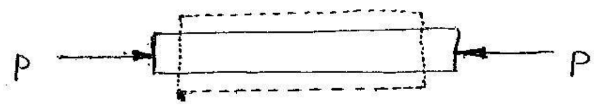
1. Tensile stress   2. Compressive stress

Sometimes a body is pulled outwards by two equal and opposite forces and tends to extend as shown in the fig below:



This stress induced is called tensile stress and coressponding force is called tensile force.

Sometimes, a body is pushed inwards by two equal and opposite forces and the body tends to shorten its length as shown below



The stress induced is called compressive stress and the coresspondIng force is called compressive force.

Every member of a truss is a two force member i.e. only two forces can act on a member which can be either tensile or compressive.

ANALYSIS OF A TRUSS :

The analysis of a truss involves the determination of :

(i) the reaction at the supports.

(ii) the internal stresses induced in the members due to external loading.

While making such an analysis following assumptions are made:

(i) The truss is a perfect one and statically determinate.

(ii) All the members comprising the truss are rigid and lie in the same plane.

(iii) The members are slender and of uniform cross sectio

(iv) The external loads and reactions act at the joints only.

(v) The weight of members are negligibly small unless otherwise mentioned.

vi) The joints of a simple truss are assumed to be pin connections and frictionless.

(vii) The members of a truss are straight two force members with the forces acting collinear with the centre line of the members.

The stresses in the members of a truss are generally worked out by adopting the following methods:

1. Analytical method
   - method of joints
   - method of sections.
2. Graphical method.

* Method of Joints

Here every joint is treated separately as a free body in equilibrium, i.e., sum of all the vertical forces as well as horizontal forces acting on the joint is equated to zero.

$$\Sigma F_v = 0 \quad \text{and} \quad \Sigma F_h = 0$$

The steps involved are:

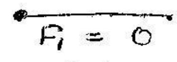
- All the pin joints are labelled.
- A free body diagram of entire truss is drawn and reaction at the supports is determined using the conditions of equilibrium.
- Each joint is treated seperately as a free body. A certain direction of forces acting on the joint is assumed and the magnitude of forces is worked out by applying the conditions of equilibrium. If the magnitude of a particular force comes out positive, the assumption in respect of direction is correct.

However if the magnitude of the force comes out negative, actual direction of force is reversed i.e. opposite to what has been assumed.

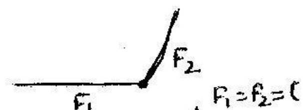
- Start is made from a joint where there are not more than two members in which the forces are unknown, and the process is repeated from one joint to another untill all the unknowns have been determined.

- A member under tension is called a tie and a member under compression is called a strut.

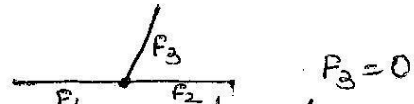
The following principles help to identify the members not subjected to any force when the truss is loaded:



(i) A single force cannot form a system in equilibrium. This implies that if there is only one force acting at a joint, then for equilibrium of this joint, this force equals zero.

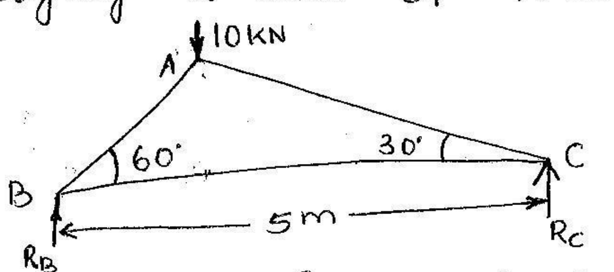


(ii) When two members meeting at a joint are not collinear and there is no external force acting at the joint, then forces in both the members are zero.



iii) When three members are meeting at an unloaded joint and out of them two are collinear, then the force in third member will be zero.

Q1. The truss shown in fig. has a span of 5 metres. It is carrying a load of 10 kN at its apex.



Find the forces in members AB, AC and BC.

Sol^n: From the geometry of the truss, we find that the load of 10kN is acting at a distance 1.25 m from the left hand support i.e. B and 3.75 m from C.

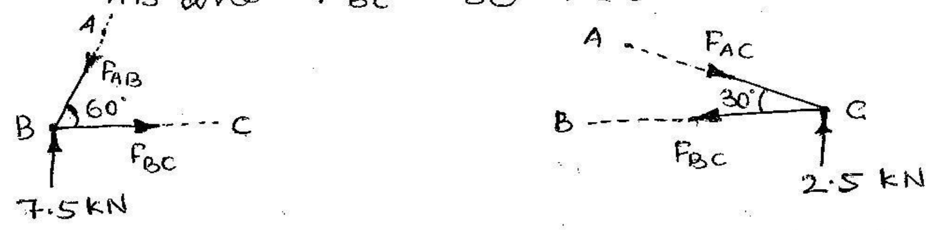
Taking moments about B and equating the same

$$R_C \times 5 = 10 \times 1.25 = 12.5$$

$$R_C = \frac{12.5}{5} = 2.5 \text{ kN}$$

As $R_B + R_C = 10$ kN $\Rightarrow R_B = 10 - R_C = 10 - 2.5 = 7.5$ k

Now, 1st of all consider joint B. Let the direction of forces F_{AB} and F_{BC} be assumed as shown below



(a) joint B (b) joint C

Resolving the forces vertically and equating the same

$$F_{AB} \sin 60° = 7.5$$

$$F_{AB} = \frac{7.5}{\sin 60°} = \frac{7.5}{0.866} = \underline{8.66 \text{ kN}} \text{ (compression)}$$

and now resolving the force horizontally and equating the same

$$F_{BC} = F_{AB} \cos 60° = 8.66 \times 0.5 = \underline{4.33} \text{ kN (Tension)}$$

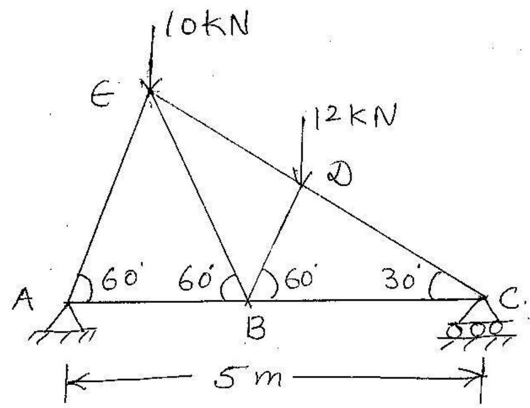
Now consider the joint C. Let the directions of forces F_{AC} and F_{BC} be assumed as shown above in (b).

Resolving the forces vertically and equating the same

$$F_{AC} \sin 30° = 2.5$$

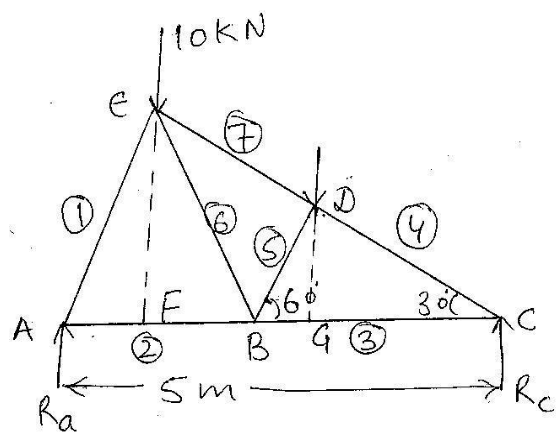
$$F_{AC} = \frac{2.5}{\sin 30°} = \frac{2.5}{0.5} = \underline{5.0 \text{ kN}} \text{ (Compression)}$$

Q-1 Determine the forces in all the members of the truss loaded and supported as shown in given fig.



Sol

Soln The reaction at the supports can be determined by considering equilibrium of the entire truss.



Since the $\triangle AEC$ is a right angle triangle with angle $AEC = 90°$ Then

$AE = AC \cos 60° = 5 \times 0.5 = 2.5m$

$CE = AC \sin 60° = 5 \times 0.866 = 4.33\ m$

Further, from the given geometry, $\triangle ABE$ is an equilateral triangle and therefore, $AB = BE = AE = 2.5\ m$

Distance of line of action of force 10kN from joint A

$AF = AE \cos 60° = 2.5 \times 0.5 = 1.25m$

Again, the triangle BDC is a right triangle with angle $BDC = 90°$. Also

$BC = AC - AB = 5 - 2.5 = 2.5m$

$BD = BC \cos 60° = 2.5 \times 0.5 = 1.25m$

Distance of line of action of force 12 KN from joint A,

$AG = AB + BG = AB + BD \cos 60^\circ$

$= 2.5 + 1.25 \times 0.5 = 3.125 m$

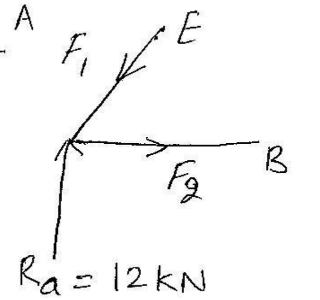
Taking moments about end A,

$R_c \times 5 = 12 \times 3.125 + 10 \times 1.25 = 50$

$\therefore R_c = \frac{50}{5} = 10 KN$ and $R_a = (10+12) - 10$

$= 12 KN$

Joint A



Eqⁿs of equilibrium can be written as

$\Sigma F_x = 0, \quad F_2 - F_1 \cos 60^\circ = 0$

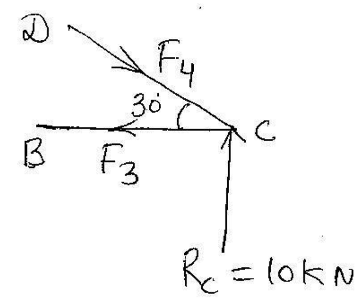
$\Sigma F_y = 0, \quad R_a - F_1 \sin 60^\circ = 0$

$\therefore F_1 = \frac{R_a}{\sin 60^\circ} = \frac{12}{0.866} = 13.85 KN$ (Compression)

$F_2 = F_1 \cos 60^\circ$

$= 13.85 \times 0.5 = 6.92 KN$ (Tension)

Joint C



Eqⁿs of equilibrium :-

$\Sigma F_x = 0 \quad F_4 \cos 30^\circ - F_3 = 0$

$\Sigma F_y = 0 \quad R_c - F_4 \sin 30^\circ = 0$

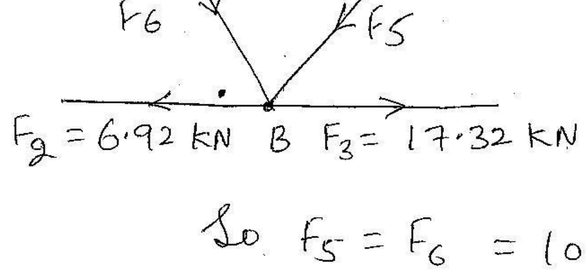
$\therefore F_4 = \frac{R_c}{\sin 30^\circ}$

$= \frac{10}{0.5} = 20 KN$ (Compression)

and $F_3 = F_4 \cos 30^\circ$

$= 20 \times 0.866 = 17.32 KN$ (Tension)

Joint B



$\Sigma F_x = 0, \quad 17.32 - 6.92 - F_5 \cos 60^\circ - F_6 \cos 60^\circ = 0$

or $(F_5 + F_6) \times 0.5 = 10.40$

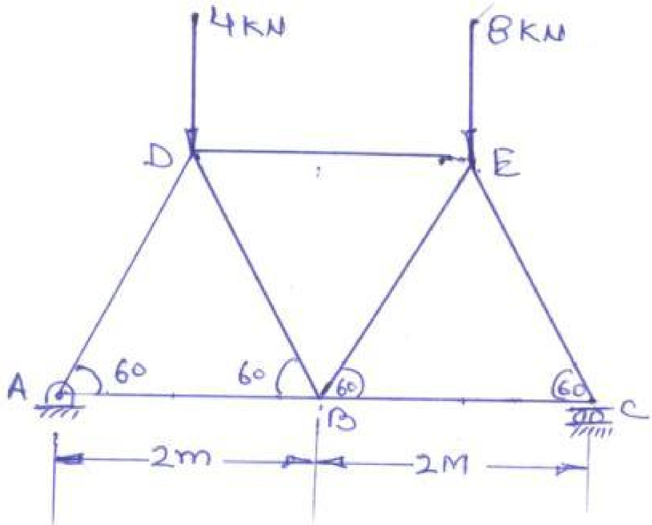
or $F_5 + F_6 = 20.80$

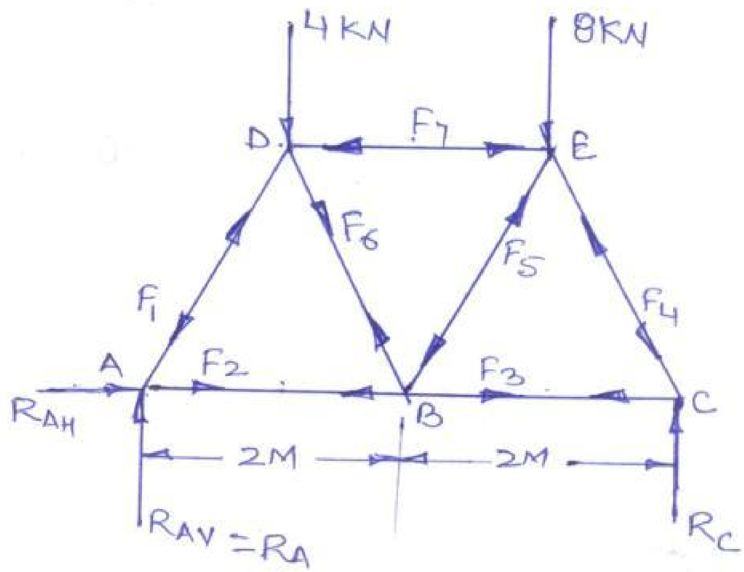
$\Sigma F_y = 0, \quad -F_5 \sin 60^\circ + F_6 \sin 60^\circ = 0$

or $F_5 = F_6$

So $F_5 = F_6 = 10.40 KN$

Determine the support reactions and the stresses in the members of the truss shown in fig below.





Assume the directions of the stresses/forces as shown in fig

As there is no any horizontal force act at p. the truss so R_{AH} become zero.

Now considering the entire truss in equlibrium and applying the equations of Equilibrium

Taking moment of all the forces about point A and Equating the same to zero

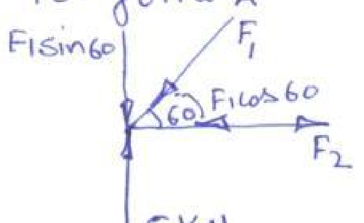
$\Sigma M_A = 0 \quad 4 \times 2 \cos 60 + 8 \times (2 + 2\cos 60) - 4 \times R_C = 0$

$4 R_C = 4 + 24 = 28 \quad$ or $\quad R_C = 7\,KN$

$\Sigma F_y = 0 \quad 12 - R_A - R_C = 0 \quad R_A = 12 - 7 = 5\,KN$

Now consider each joint as a free body separately in equilibrium and apply Equation of Equilibrium.

For joint A

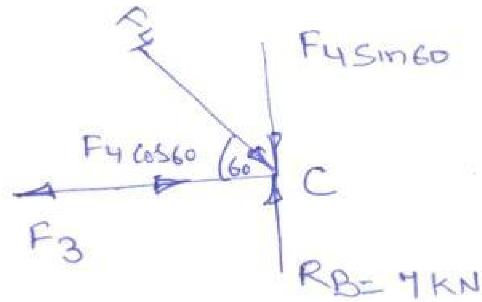


$\Sigma F_x = 0 \quad F_1 \cos 60 = F_2$

$\Sigma F_y = 0 \quad F_1 \sin 60 = 5 \quad$ or $\quad F_1 = \frac{5}{\sin 60} = \frac{5}{0.866} = 5.77\,KN\,(C)$

$F_2 = F_1 \cos 60 = 5.77 \cos 60 = 2.88\,KN\,(T)$

For joint C



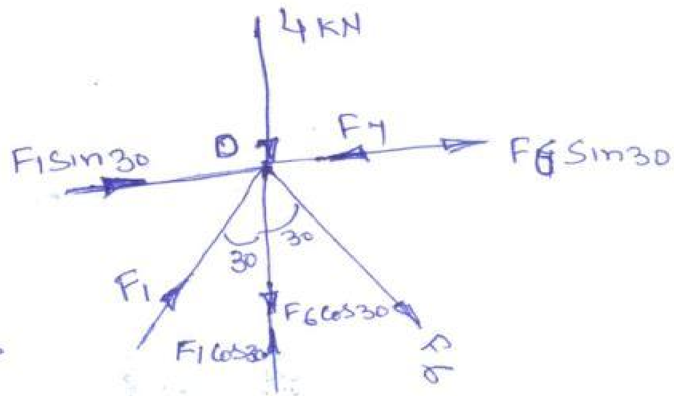
$\Sigma F_x = 0 \qquad F_3 = F_4 \cos 60$

$\Sigma Y = 0$ gives $F_4 \sin 60 = 7$

$$F_4 = \frac{7}{\sin 60} = 8.08\,\text{KN (C)}$$

$$F_3 = F_4 \cos 60 = 8.08 \cos 60 = 4.04\,\text{KN (T)}$$

For joint D



$\Sigma F_x = 0$

gives $F_7 = F_6 \sin 30 + F_1 \sin 30$

$F_7 = F_6 \sin 30 + 5.77 \sin 30$

$F_7 = 1.151 \sin 30 + 5.77 \sin 30.$

$= 3.46\,\text{KN (C)}$

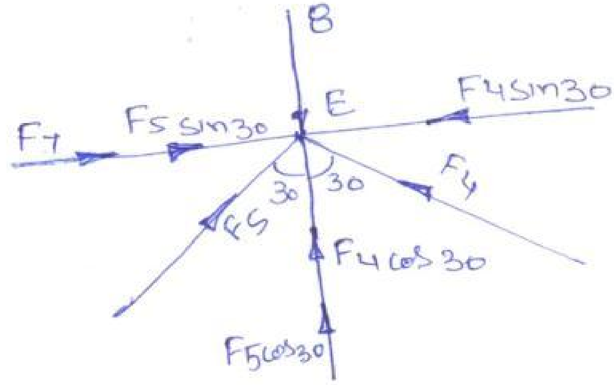
$\Sigma F_y = 0$ gives

$F_6 \cos 30 + 4 = F_1 \cos 30$

$F_6 \cos 30 = 5.77 \cos 30 - 4 = 0.9969$

$$F_6 = \frac{0.9969}{\cos 30} = 1.151\text{N (T)}$$

Joint E.



$\Sigma F_x = 0$ gives

$F_4 \sin 30 = F_7 + F_5 \sin 30$

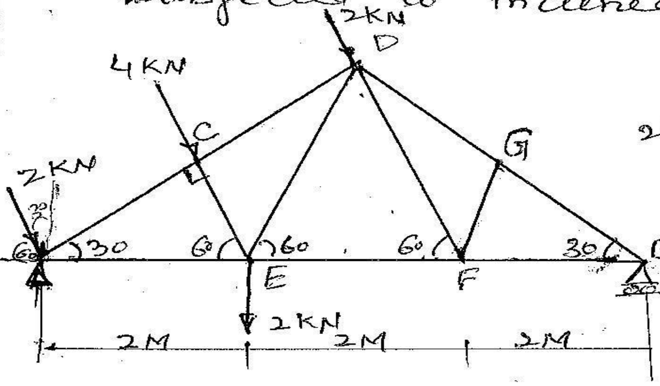
$F_5 \sin 30 = F_4 \sin 30 - F_7$

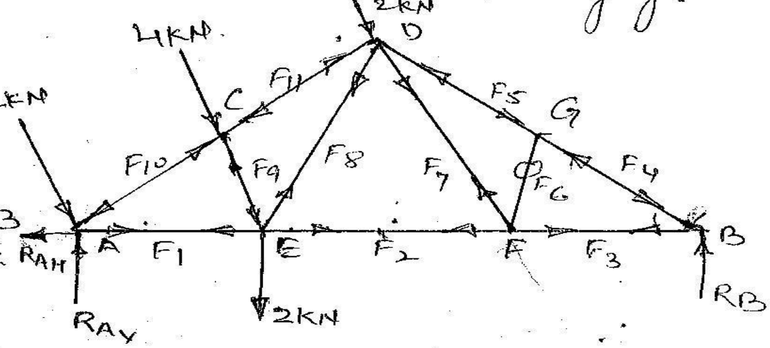
$= 8.08 \sin 30 - 3.46$

$F_5 \sin 30 = 4.04 - 3.46 = .58$

$$F_5 = \frac{0.58}{\sin 30} = 1.16\,\text{KN (C)}$$

Findout the forces in each member of a truss subjected to inclined loads as shown in fig.





Considering the Equlibrium of entire truss.

Length $AC = 2\cos 30 = 1.732M$

$AD = 4\cos 30 = 3.464M$

Taking moment of all the forces about pt A

$\Sigma M_A = 0 \quad 4 \times 1.732 + 2 \times 3.464 + 2 \times 2 = 6R_B \qquad R_B = 2.98$

Total vertical Component of inclined forces

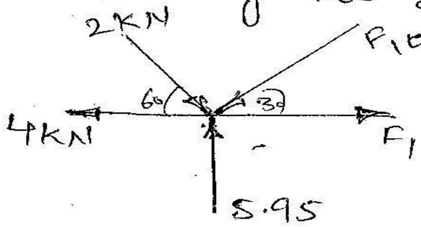
$(2+4+2)\sin 60 + 2 = 8.93\,KN$

Total Horizontal component of inclined forces

$(2+4+2)\cos 60 = 4\,KN$

Now $R_{AH} = 4\,KN$ & $R_{AY} = 8.93 - 2.98 = 5.95\,KN$

Considering the Equlibrium of joint A



$\Sigma F_x = 0 \quad F_1 - 4 - F_{10}\cos 30 + 2\cos 60 = 0$

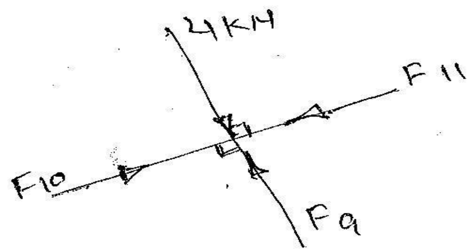
$F_1 - F_{10}\cos 30 = 4 - 1.0 = 3.00\,KN$

$\Sigma F_y = F_{10}\sin 30 = 5.95 - 1.732 = 4.2179$

$F_{10} = \dfrac{4.2179}{\sin 30} = 8.43\,KN\,(C)$

$F_1 = 3 + F_{10}\cos 30 = 10.30\,KN\,(T)$

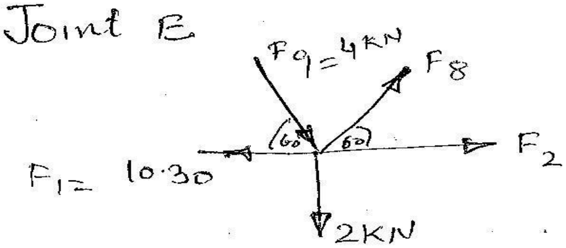
Joint C



$\Sigma F_x = 0 \quad F_{11} = F_{10} = 8.43\,KN\,(C)$

$F_9 = 4\,KN\,(C)$

Joint E



$\Sigma F_x = 0$

$-10.30 + 4\cos 60 + F_8 \cos 60 + F_2 = 0$

$F_2 = 10.30 - 4\cos 60 - 6.30 \cos 60$

$= 10.30 - 2 - 3.15 = 5.15(T)$

$\Sigma F_y = 0$

$2 + 4\sin 60 - F_8 \sin 60 = 0$

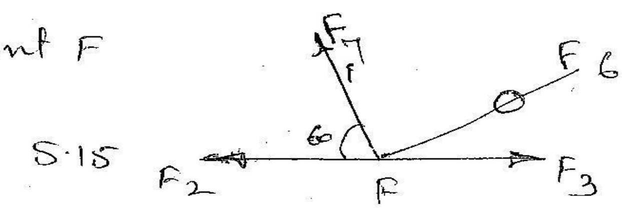
$F_8 \sin 60 = 2 + 3.46 = 5.46$

$F_8 = \dfrac{5.46}{\sin 60} = 6.30\,KN(T)$

there is no loads

At joint G, three forces F_4, F_5, F_6 are meeting out of which F_4 & F_5 are collinear Hence $F_6 = 0$ and $F_4 = F_5$

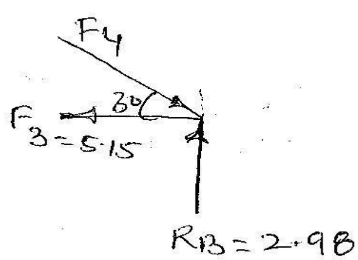
Joint F



$\Sigma F_y = 0 \quad F_7 \sin 60 = 0 \quad$ Hence $F_7 = 0$

$\Sigma F_x = 0 \quad F_2 = F_3 = 5.15(T)$

Joint B



$F_4 \cos 30 = 5.15$

$F_4 = \dfrac{5.15}{\cos 30} = 5.96\,KN\,(C)$

Joint G

$F_4 = F_5 = 5.96\,kN\,(C)$

* Method of Sections :

The various steps involved are

(i) The truss is split into two parts by passing an imaginary section.

ii) The imaginary section has to be such that it does not cut more than three members in which the forces are to be determined.

iii) The conditions of equilibrium

$$\Sigma F_x = 0 ; \quad \Sigma F_y = 0 \quad \text{and} \quad \Sigma M = 0$$

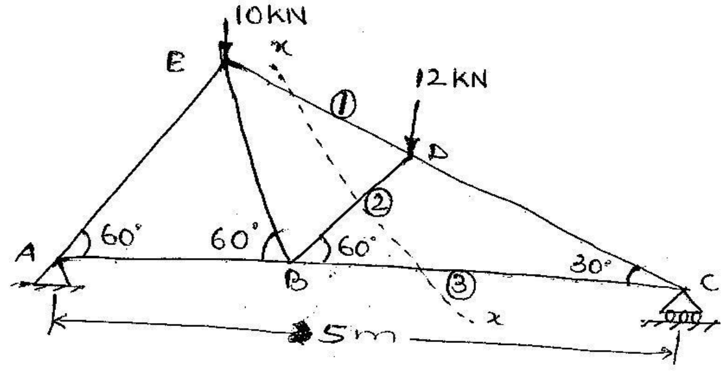
are applied for the one part of the truss and the unknown force in the member is determined.

iv) While considering equilibrium, the nature of force in any member is chosen arbitrarily to be tensile or compressive.

If the magnitude of a particular force comes out positive, the assumption in respect of its direction is correct. However, if magnitude of the force comes out negative, the actual direction of the force is opposite to that what has been assumed.

The method of sections is particularly convenient when the forces in a few members of the frame is required to be worked out.

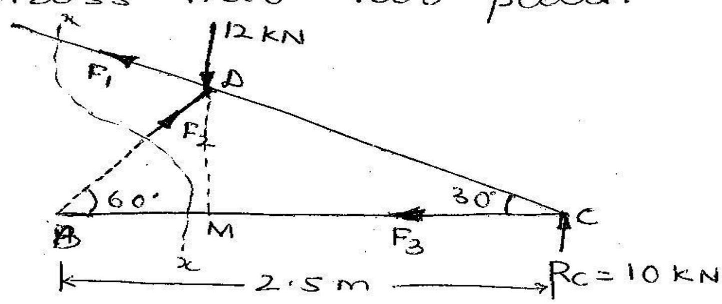
Q2. Determine the forces in the members marked (1), (2) and (3) of the truss loaded and supported as shown in fig.



Soln: First of all the reactions at the supports are determined.

$R_a = 12$ kN and $R_C = 10$ kN.

Let a section x-x be passed in such a way that it cuts the members marked (1), (2) and (3) and divides the truss into two parts.



$CD = BC \cos 30° = 2.5 \times 0.866 = 2.165$ m.

$BD = BC \sin 30° = 2.5 \times 0.5 = 1.25$ m

Under equilibrium conditions of right part of truss

(i) taking moments about C,

$$F_2 \times CD - 12 \times MC = 0$$

$$\therefore \quad F_2 = 12 \times \frac{MC}{CD} = 12 \times \frac{CD \cos 30°}{CD}$$

$$= 12 \times 0.866 = \underline{\underline{10.39 \text{ kN}}} \text{ (compression)}$$

ii) Taking moments about B,

$$F_1 \times BD + R_C \times BC - 12 \times BM = 0$$

or $\quad F_1 \times 2.5 \sin 30° + 10 \times 2.5 - 12 \times 1.25 \cos 60° = 0$

or $\quad 1.25 F_1 + 25 - 7.5 = 0$

$$\therefore \quad F_1 = \frac{7.5 - 25}{1.25} = \underline{\underline{-14 \text{ kN}}}$$

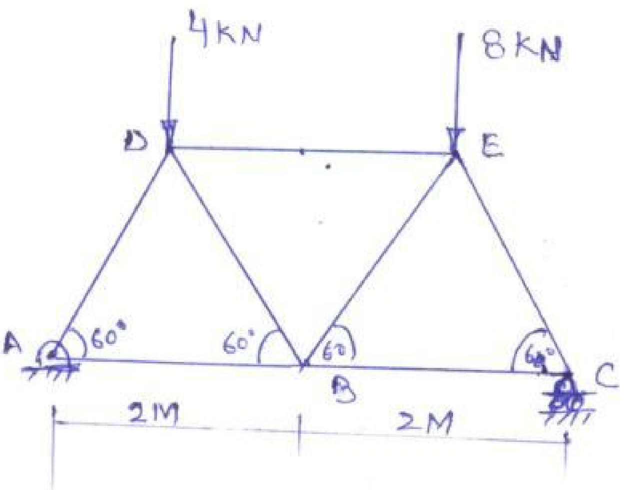
Negative sign indicates that F_1 is compressive.

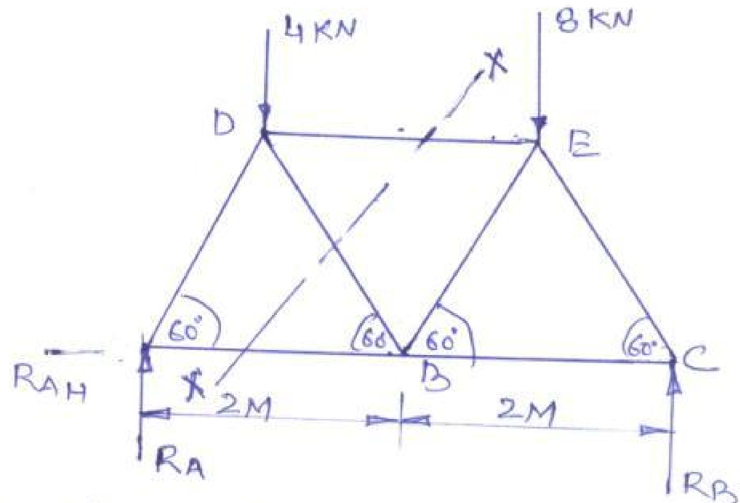
iii) Taking moments about D

$F_3 \times DM - R_C \times MC = 0 \Rightarrow F_3 = R_C \times \frac{CD \cos 30°}{CD \sin 30°}$

$= \underline{\underline{17.32}}$ kN (tension)

Determine the forces in the members BD, DE and AB of the truss shown in fig. below.





Firsly find out Reactions at points A and C by considering the equilibrium of entire truss by applying the conditions of Equilibrium. i.e. $\Sigma F_x = 0$ $\Sigma F_Y = 0$ $\Sigma M = 0$

Taking moment about Point A

$\Sigma M_A = 0$ gives us

$$4 \times 2\cos 60 + 8 \times (2 + 2\cos 60) - R_C \times 4 = 0$$

$$R_C = \frac{4 + 24}{4} = \frac{28}{4} = 7\text{ KN}$$

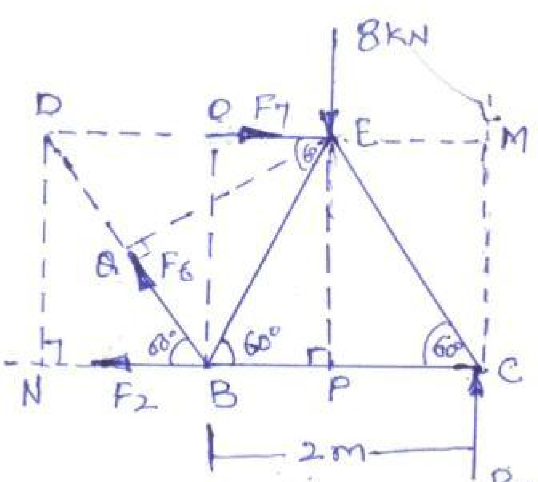
From $\Sigma F_Y = 0$ we get

$$12 - R_A - R_B = 0$$

$$R_A = 12 - 7 = 5\text{ KN}$$

Since there is no horizontal force acting on the truss hence Reaction at pt A R_A becomes zero

Now take a section throug the members in whice the forces are to be determine as shown in fig



Now considering the right Part of the truss in Equilibrium.

Taking Moment of all the forces about the joint D.

$\Sigma M_D = 0$ gives

$$F_2 \times DN + 8 \times DE - R_C \times DM = 0 \qquad (DN = 2\sin 60)$$

$$F_2 \times 2\sin 60 + 8 \times 2 - 7 \times 3 = 0$$

$$F_2 = \frac{21-16}{[illegible]} = \frac{5}{[illegible]} = 2.88\text{KN}(T)$$

Now taking moment of all the forces about joint B and applying the condition of Equilibrium $\Sigma M_B = 0$ gives

$$F_7 \times BO + 8 \times BP - R_C \times BC = 0$$

$BO = 2\sin 60$ $BP = 2\cos 60$ $BC = 2$

$$\therefore F_7 \times 2\sin 60 + 8 \times 2\cos 60 - 7 \times 2 = 0$$

$$F_7 = \frac{7 \times 2 - 8 \times 2\cos 60}{2\sin 60} = \frac{14-8}{1.732} = \frac{6}{1.732} = 3.46\ \text{KN}\ (C)$$

Now taking moment of all the forces about joint E and applying the Conditions of Equilibrium $\Sigma M_E = 0$

$$F_2 \times PE + F_6 \times QE - R_C \times EM = 0$$

$PE = 2\sin 60$ $QE = 2\sin 60$ $EM = PC = 2\cos 60$

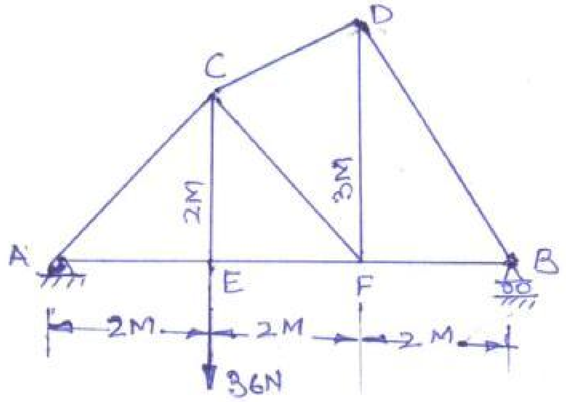
$F_2 = 2.88\ \text{KN}$ $R_C = 7$

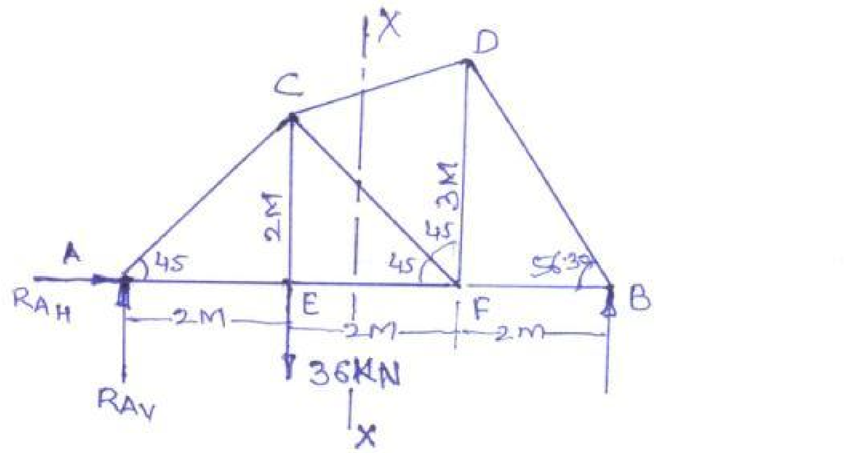
$$2.88 \times 2\sin 60 + F_6 \times 2\sin 60 - 7 \times 2\cos 60 = 0$$

$$F_6 \times 2\sin 60 = 7 \times 2\cos 60 - 2.88 \times 2\sin 60$$

$$F6 = \frac{7 \times 2\cos 60 - 2.88 \times 2\sin 60}{2\sin 60} = \frac{7 - 4.998}{1.732} = 1.155\ \text{KN}(T)$$

A Plane truss is loaded as shown in fig. below. Find the axil forces in Members CD, CF and EF.





Since No any horizontal load is acting on the truss Hence horizontal reaction R_{AH} at Point A becomes zero and $R_{AV} = R_A$

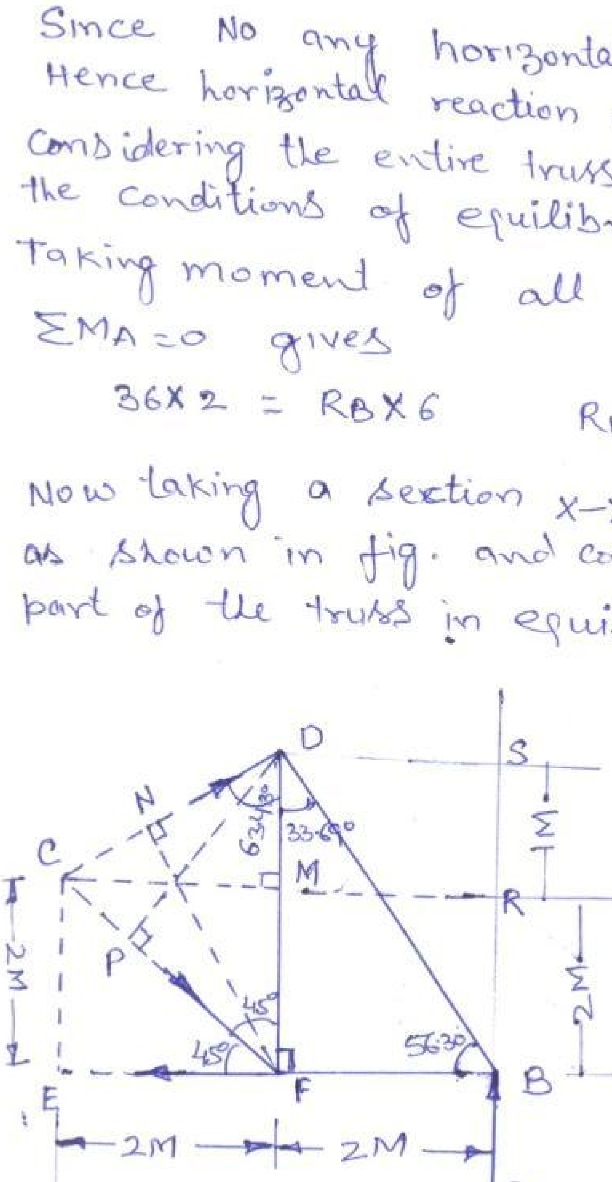
Considering the entire truss in equilibrium and applying the conditions of equilibrium $\Sigma M = 0$

Taking moment of all the forces about pt A and equating it to zero

$\Sigma M_A = 0$ gives

$36 \times 2 = R_B \times 6$ $\quad R_B = \frac{36 \times 2}{6} = 12\ KN$ $\quad R_A = 36 - 12 = 24\ KN$

Now taking a section X–X through the members CD, CF & EF as shown in fig. and considering the equilibrium of Right part of the truss in equilibrium



IN Δ FDB $\tan\theta = \frac{DF}{FB} = \frac{3}{2} =$

$\theta = \tan^{-1}\left(\frac{3}{2}\right) = 56.30°$

and $\angle FDB = 90 - 56.30 = 33.69°$

IN Δ CEF $\quad \angle CFE = 45°$

Hence angle CFD in Δ FCD is also equal to 45°

In Δ CMD $\angle CDM$ Can be calculated as $\tan\angle CDM = \frac{P}{B} = \frac{2}{1}$

Lengths

FN in Δ FND is equal to $3 \sin 63.43° = 2.683M$

PD IN Δ FPD $= 3 \sin 45 = 2.121m$

Now taking moment of all the forces about pt F and equating to zero. we ge

$\Sigma M_F = 0$

$F_{CD} \times FN = R_B \times 2$

$F_{CD} \times 2.683 = 12 \times 2$ or $F_{CD} = \frac{24}{2.683} = 8.94 \text{ KN } (C)$

Moment about pt C

$\Sigma M_C = 0$ gives us

$F_{EF} \times CE - R_B \times CR = 0$ $(CR = EB = 4M, \quad CE = 2M)$

$F_{EF} \times 2 = 12 \times 4$

$F_{EF} = \frac{12 \times 4}{2} = 24 \text{ KN } (C)$

Moment about Pt D

$\Sigma M_D = 0$ gives us

$F_{EF} \times FD - F_{CF} \times PD - R_B \times DS = 0$

$\left(\begin{array}{l} FD = 3M \text{ Given} \\ PD = 3 \sin 45 = 2.121M \\ DS = FB = 2M \end{array} \right)$

$F_{EF} \times 3.0 - F_{CF} \times 2.121 - 12 \times 2 = 0$

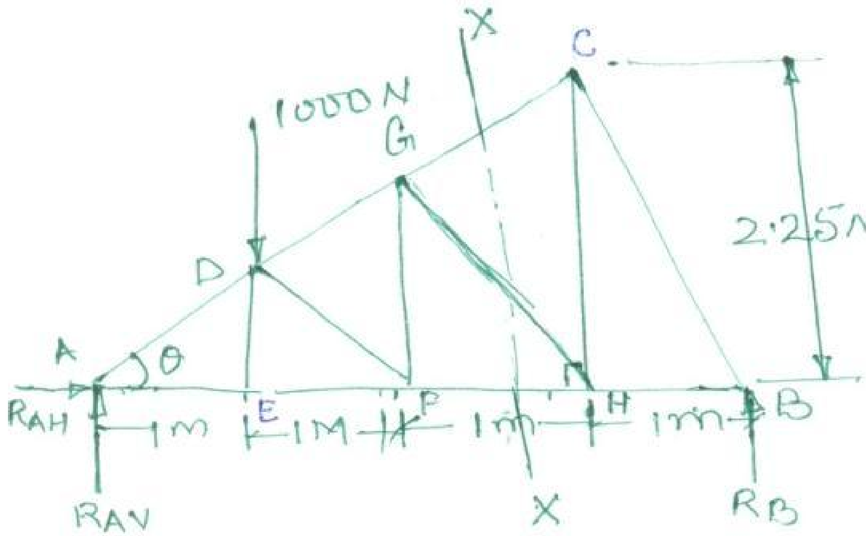
$F_{CF} \times 2.121 = 24 \times 3 - 12 \times 2$

$F_{CF} = \frac{72 - 24}{2.121} = \frac{48}{2.121} = 22.63 \text{ KN } (C)$

ASSIGNMENT NO. 2

Q.No.2.

The given question,



In the given fig there is no external horizontal load is acting Hence the horizontal Part of reaction at joint A i.e R_{AH} becomes zero.

Angle θ In Δ ACH can be calculated as

$\tan\theta = \frac{P}{B} = \frac{2.25}{3} = .75$ $\theta = \tan^{-1}(.75) = 36.86°$

also the angle EFD = θ. The angle $\angle ADE$ in triangle ADE $= 180 - (90 - 36.86) = 53.14°$

Similarly the angle $\angle ACH$ in Δ ACH is also equal to 53.14°

Now for finding out Reactions conside the entire truss in equilibrium and appling the Equation of Equilibrium.

Taking Moment of all the forces about point A

$\Sigma M_A = 0$ which gives us

$1000 \times 1 - R_B \times 4 = 0$ $R_B = \frac{1000}{4} = 250\,N$

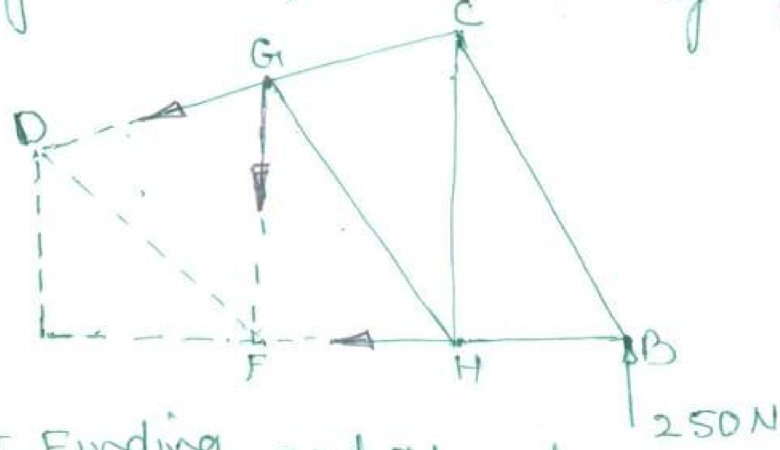
Now $\Sigma F_Y = 0$ Which gives us

$1000 - R_{AV} - R_B = 0$

Now by method of sections

Now cut the fig by a plane X-X as shown in fig in two separate parts

Considering the equilibrium of Right Part.



Now For Finding out the forces

$\Sigma M_G = 0$ gives us

$$F_{FH} \times 1.5 = \frac{250 \times 2}{1.5} = 333.33N$$

$\Sigma M_F = 0$

$$F_{DG} \times GF \sin 53.14 = 250 \times 2$$

$$F_{DG} = \frac{250 \times 2}{1.5 \sin 53.14} = 416.66$$

$\Sigma M_A = 0$

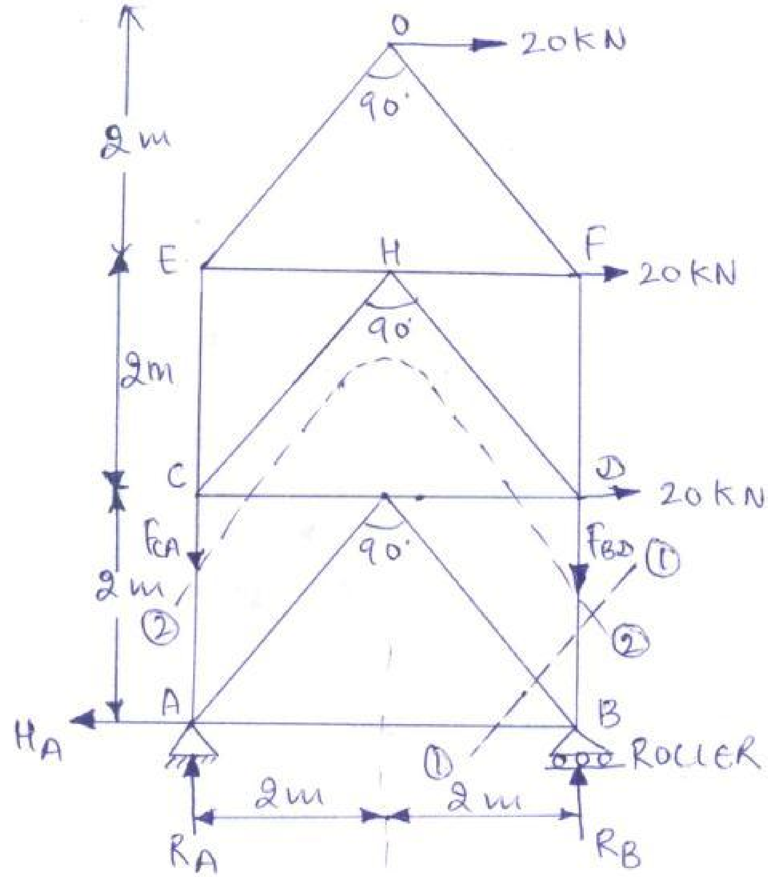
$\Sigma M_C = 0$

$$F_{FH} \times 2.25 = F_{FG} \times 1 + 250 \times 1$$

$$333.33 \times 2.25 = F_{FG} + 250$$

$$F_{FG} = 750 - 250 = 500N$$

For the Frame shown in given Fig. find the forces in the members BD, BG, GA, AC & AB of the bottom bay only. State their nature.



Taking moments of all forces about A

$R_B \times 4 = 20 \times 2 + 20 \times 4 + 20 \times 6 = 240$

$\therefore R_B = \frac{240}{4} = 60 \text{ KN } (\uparrow)$

Now R_A = Total vertical loads $- R_B$

$= 0 - 60 = -60 \text{ KN}$

-ve sign means, R_A is acting downwards

$\therefore R_A = 60 \text{ KN} (\downarrow)$

& H_A = sum of all horizontal loads

$= (20 + 20 + 20) = 60 \text{ KN} (\leftarrow)$

Now Taking the moments of all forces about point C,

$20 \times 2 + 20 \times 4 + F_{BD} \times 4 = 0$

$F_{BD} = \frac{-40-80}{4} = \frac{-120}{4} = -30 \text{ KN}$

-ve sign means the force F_{BD} is compressive

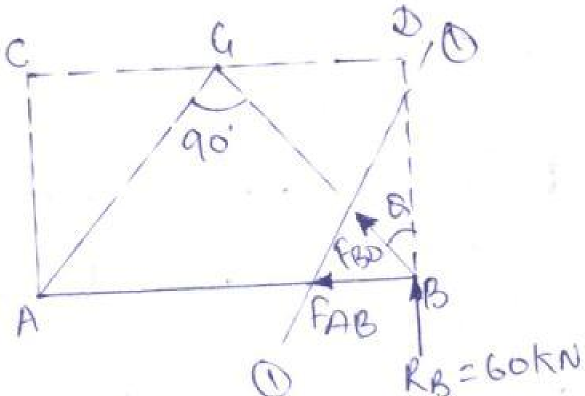
$\therefore F_{BD} = 30 kN$ (compressive)

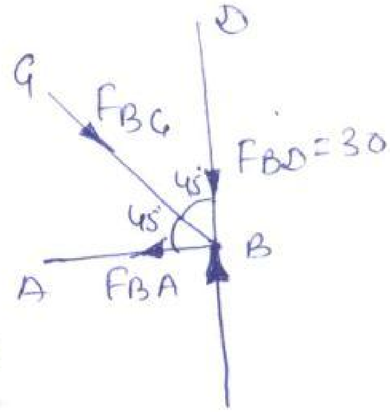
Now taking moments of all forces about point D

$$F_{CA} \times 4 = 20 \times 2 + 20 \times 4 = 120$$

$$F_{CA} = \frac{120}{4} = 30 kN \text{ (Tensile)}$$

Joint B





Resolving forces vertically,

$$F_{BG} \cos 45° + F_{BD} = R_B$$

$$F_{BG} \times \cos 45° + 30 = 60$$

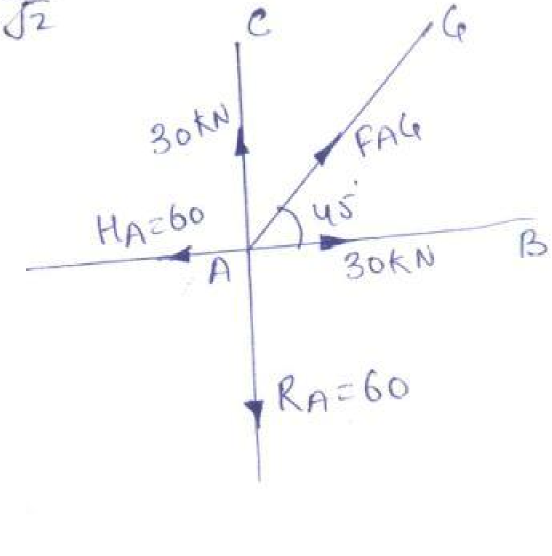
$$F_{BG} = \frac{60-30}{\cos 45°} = \frac{30}{1/\sqrt{2}} = 30\sqrt{2} \text{ Compressive}$$

Resolving horizontally, we get

$$F_{BA} = F_{BG} \sin 45°$$

$$= 30 \times \sqrt{2} \times \frac{1}{\sqrt{2}} = 30 kN \text{ (Tensile)}$$

Joint A



Resolving horizontally,

$$60 = 30 + F_{AG} \cos 45°$$

$$F_{AG} = \frac{60-30}{\cos 45°} = \frac{30}{1/\sqrt{2}}$$

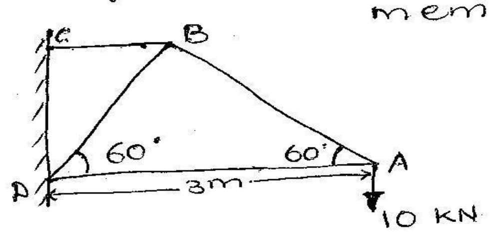
$$= 30 \times \sqrt{2} \text{ (Tensile)}$$

Cantilever Trusses :

A truss, which is connected to a wall or a column at one end, and free at the other is known as a cantilever truss.

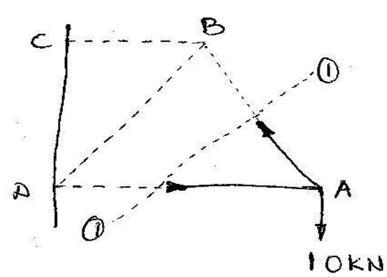
In case of cantilever trusses, determination of support reaction is not essential, as we can start the calculation work from the free end of the cantilever.

Q3. A cantilever truss of 3 m span is loaded as shown in fig. Find the forces in the various members of truss.

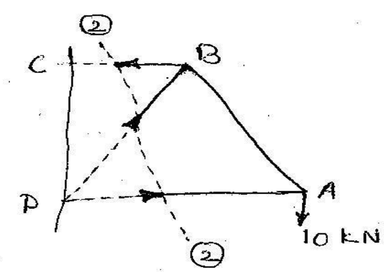


Sol^n: by method of sections.

First of all, pass section (1-1) cutting the truss through the members AB and AD. Now consider equilibrium of right part of truss.



(a) section (1-1)



(b) section (2-2)

Taking moments about point D

$$F_{AB} \times 3 \sin 60^\circ = 10 \times 3$$

$$\Rightarrow F_{AB} = \frac{30}{3 \sin 60^\circ} = \underline{\underline{11.5 \text{ kN}}} \text{ (Tension)}$$

Taking moments about point B and equating the same,

$$F_{AD} \times 3 \sin 60^\circ = 10 \times 1.5$$

$$\Rightarrow F_{AD} = \frac{15}{3 \times 0.866} = \underline{\underline{5.75 \text{ kN}}} \text{ (Compression)}$$

Now pass section (2-2) cutting the truss through the members BC, BD and AD. Now consider the equilibrium of right part of truss.

Taking moments about joint D,

$$F_{BC} \times 3 \sin 60^\circ = 10 \times 3$$

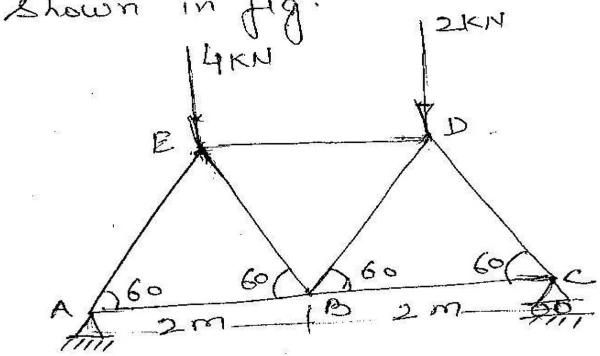
$$F_{BC} = \frac{10}{0.866} = \underline{\underline{11.5 \text{ kN}}} \text{ (Tension)}$$

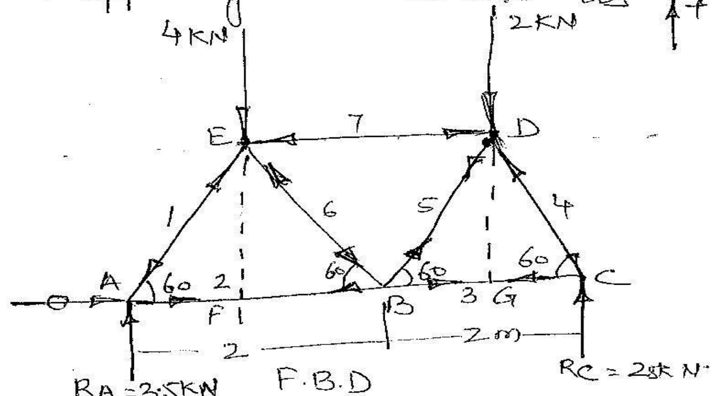
Now taking moments about joint C,

$$F_{BD} \times 1.5 \sin 60^\circ = (10 \times 3) - F_{AD} \times 3 \sin 60^\circ$$

$$F_{BD} = \frac{15}{1.5 \times 0.866} = \underline{\underline{11.5 \text{ kN}}} \text{ (Compression}$$

(1)
Determine the reactions and the forces in each member of a simple triangular truss supporting two loads as shown in fig.





Reaction at Hing support Horizontal & vertical $R_{AH} = 0$ Horizontal load is not there

Distance of line of action of 4KN load from pt A

$AF = AE \cos 60 = 2 \times .5 = 1m.$

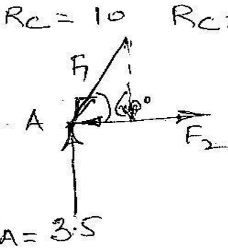
Distance of line of action of 2KN load from pt A $= AG = AB + BD \cos 60 = 2 + 1 = 3m.$

Now $M_A = 4 \times 1 + 2 \times 3 - R_C \times 4 = 0$ $\quad 4R_C = 10 \quad R_C = 2.5KN$

$R_A = (4+2) - 2.5 = 3.5KN$

Considering the F.B.D of joint A

equations of Equilibrium



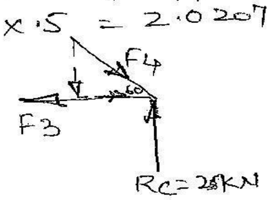
$\Sigma F_H = 0 \quad F_2 - F_1 \cos 60 = 0$

$\Sigma F_V = 0 \quad R_A - F_1 \sin 60 = 0$

$F_1 = \dfrac{R_A}{\sin 60} = 4.041$ KN (Comp)

$F_2 = F_1 \cos 60 = 4.041 \times .5 = 2.0207$ KN (Ten)

F.B.D of joint C

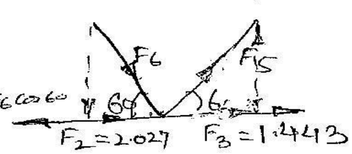


$\Sigma F_H = 0 \quad F_4 \cos 60 - F_3 = 0$

$\Sigma F_V = 0 \quad R_C - F_4 \sin 60 = 0 \qquad F_4 = \dfrac{R_C}{\sin 60} = \dfrac{2.5}{.866} = 2.886$KN (Comp)

$F_3 = 2.886 \times 0.5 = 1.443$ KN (Ten)

Considering joint B.



$\Sigma F_H = 0,\ 1.443 - 2.0207 + F_5 \cos 60 + F_6 \cos 60 = 0$

or $(F_5 + F_6) \times .5 = 0.58$

$F_5 + F_6 = 1.16$

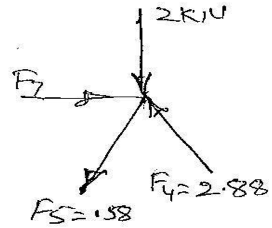
$\Sigma F_V = 0 \quad F_5 \sin 60 - F_6 \sin 60 = 0$

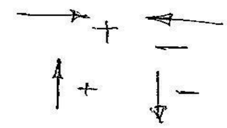
$F_5 = F_6$

$F_5 = F_6 = \dfrac{1.16}{2} = .58$KN

$F_5 = .58$ KN (Ten) $\quad F_6 = .58$KN (Com)

Joint D



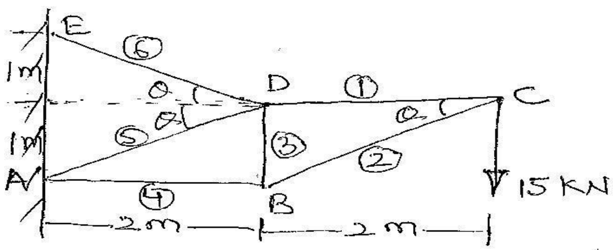


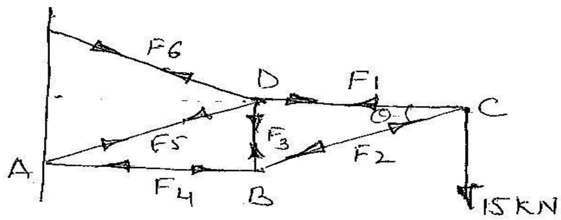
$\Sigma F_H = 0$

$F_7 - 0.58 \cos 60 - 2.88 \cos 60 = 0$

$F_7 = 0.58 \times 0.5 + 2.88 \times 0.5 = 0.29 + 1.44 = 1.73$ KN (Com)

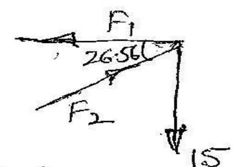
Determine the forces in various members of a cantilever truss loaded & supported as shown in fig





$\tan \theta = \frac{P}{B} = \frac{1}{2} = .5 \quad \theta = \tan^{-1} .5 = 26.56°$

Considering the F.B.D of joint C



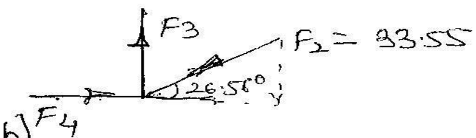
$\Sigma F_H = 0, F_2 \cos 26.56 - F_1 = 0$

$\Sigma F_V = 0 \quad F_2 \sin 26.56 - 15 = 0$

$F_2 = \frac{15}{\sin 26.56} = 33.55$ KN (Comp)

$F_1 = F_2 \cos 26.56 = 33.55 \cos 26.56 = 33.55 \times .8945 = 30$ KN (Ten)

Considering the F.B.D. of joint B



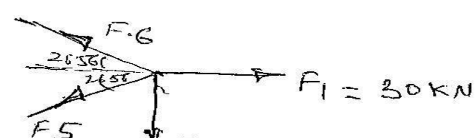
$\Sigma F_H = 0 \quad F_4 - 33.55 \cos 26.56 = 0$

$F_4 = 33.55 \times .8945 = 30$ KN (Comp)

$\Sigma F_V = 0 \quad F_3 - 33.55 \sin 26.56° = 0$

$F_3 = 33.55 \times .447 = 15$ KN (Ten)

Considering the F.B.D of joint D



$\Sigma F_H = 0 \quad F_1 - F_5 \cos 26.56 - F_6 \cos 26.56 = 0$

$30 = F_5 \cos 26.56 + F_6 \cos 26.56$ or $F_5 + F_6 = \frac{30}{\cos 26.56} = \frac{30}{.894} = 33.55$ KN

$\Sigma F_V = F_6 \sin 26.56 - F_5 \sin 26.56 - 15 = 0$

$F_6 - F_5 = \frac{15}{\sin 26.56} = \frac{15}{.447} = 33.55$

$F_6 - \cancel{F_5} = 33.55$

$F_6 + \cancel{F_5} = 33.55$

$2F_6 = 67.10 \quad F_6 = \frac{67.10}{2} = 33.55$ KN (Ten) hence $F_5 = 0$